An energy budget

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect work, motion, and heating. State a conclusion that fits the calculation's limits.

SUCCESS CRITERIA - I can...

1. Keep all terms in consistent energy units.
2. Distinguish mechanical-energy change from total-energy loss.
3. State what you treat as true when you account for the energy.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "energy budget" and "system" to explain your answer. Sentence starters are optional.

energy budget: an account of all energy stored and transferred

system: the object or objects chosen for analysis

thermal energy: internal energy associated with microscopic motion/interactions

01 Warm-up: bring back an earlier idea

A **0.3 kg** object moving at **5 m/s** stops through friction over **0.7 m**. **Find energy transformed** and average opposing-force magnitude.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $E_i + E_{in} = E_f + E_{out}$.
i/f = initial/final stored energy; in/out = transfers across the chosen system boundary. Account for every form.

Worked reasoning: An input of **100 J** gives **60 J** useful motion, **25 J** thermal transfer and **15 J** sound/other transfers. The account balances because **60+25+15=100 J**.

Your turn: Draw proportional energy arrows and **calculate the useful fraction**.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Initial **Transfers** **Final**

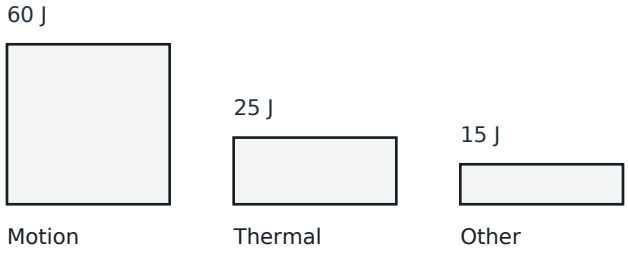
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A modeled system starts with **100 J**. It ends with **60 J** motion energy, **25 J** thermal energy and **15 J** sound/deformation. **Draw an energy account** and **check the total**.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A simulated **1000 kg** probe has K = **8 × 10⁶ J** and gravitational U = **-6 × 10⁶ J**, with U = **0** at infinity. **Find total mechanical energy**. Drag then removes **3 × 10⁶ J**. **Complete the energy account** and discuss whether the final state is bound in a calculation with only the central body's gravity acting.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

05 Explain your reasoning

What information is missing if an impact report gives only total kinetic energy and says all of it heats the ground?

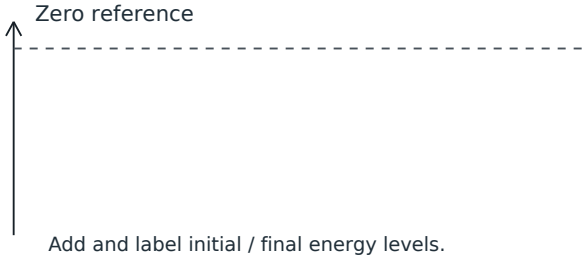
Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

With GMm=**192 J·m**, move an object from r=**4** to r=**8 m**. **Find work** by gravity using U=-GMm/r. **Label the distances or energy** levels used in your reasoning.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Add and label initial / final energy levels.

Marzano check - circle: 1 2 3

07 Exit check

A **2 kg** cart descends **3 m** from rest with g=**10**. Losses are **12 J**.
1. **Find bottom speed**.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both ... . They differ in ... because ... .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A device takes **600 W** for **5 s** and raises a **10 kg** load **12 m** using g=**10**.
1. **Find useful work**, efficiency and remaining energy.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both ... . They differ in ... because ... .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3